# Water Purification

# (Intelligent Alkaline Water Generation Module – IoT & AI/ML)

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Project Proposal Report

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## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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| Mr. S.M.B. Harshanath | |  |

## ABSTRACT (Maximum 300 words)

Alkaline water, characterized by a higher pH (typically 8-9) and the presence of essential minerals like calcium, magnesium, and potassium, has gained significant popularity for its purported health benefits, including improved hydration and acid-neutralizing properties. However, the production of consistent, safe, and truly beneficial alkaline water is a delicate balancing act. Existing commercial systems, whether using electrolysis or static mineral cartridges, often operate "open-loop," lacking the intelligence to adapt to source water variations, leading to inconsistent pH levels, improper mineralization, or even water that fails to meet safety standards.

This research proposes the development of an **Intelligent Alkaline Water Generation Module** that closes this loop by integrating **Internet of Things (IoT)** sensing with **Artificial Intelligence (AI) and Machine Learning (ML)**. The system will function as an adaptive, closed-loop controller for water mineralization. A network of precision sensors will continuously monitor key water quality parameters in real-time, including pH, Total Dissolved Solids (TDS), and individual mineral ion concentrations (where feasible).

This real-time data stream will feed into a trained machine learning model, which acts as the "brain" of the system. The model will analyze the current water state and the desired target profile (e.g., pH 8.5, specific mineral levels) to calculate the precise dosage of minerals required. It will then command an automated dosing mechanism to release the exact amounts of mineral solutions (e.g., calcium, magnesium, potassium) into the water stream. This continuous feedback loop of sense-analyze-dose ensures that the output water consistently meets the target alkaline profile, regardless of fluctuations in the input water quality. The result is a smart, reliable, and efficient system that delivers health-appropriate, mineral-balanced alkaline water on demand, setting a new standard for home water treatment.

*Keywords: Alkaline Water, Water Purification, IoT (Internet of Things), Artificial Intelligence (AI), Machine Learning, Mineral Balancing, pH Monitoring, Total Dissolved Solids (TDS), Automated Dosing, Smart Water System*

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## LIST OF ABBREVIATIONS

| Abbreviation | Description |
|---|---|
| **AI** | Artificial Intelligence |
| **ML** | Machine Learning |
| **IoT** | Internet of Things |
| **TDS** | Total Dissolved Solids (a measure of combined dissolved minerals and salts) |
| **pH** | Potential of Hydrogen (a measure of acidity or alkalinity) |
| **ANN** | Artificial Neural Network |
| **SVR** | Support Vector Regression (a type of ML for predicting continuous values) |
| **PID** | Proportional-Integral-Derivative (a control loop mechanism) |
| **ESP32** | A low-cost, low-power system on a chip microcontroller with integrated Wi-Fi |
| **MCU** | Microcontroller Unit |
| **SSR** | Solid State Relay (for controlling pumps/dosing mechanisms) |
| **UI/UX** | User Interface / User Experience |
| **API** | Application Programming Interface |
| **JSON** | JavaScript Object Notation (a lightweight data-interchange format) |
| **BOM** | Bill of Materials |

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# 1. INTRODUCTION

## 1.1 Background and Context

The quality of drinking water is not solely defined by the absence of contaminants. For many, it also encompasses the presence of beneficial elements and specific properties, such as alkalinity. Alkaline water, typically with a pH between 8 and 9, is often produced by adding minerals like calcium, magnesium, and potassium. Proponents, supported by some studies (Heil, 2012), suggest that it can act as a buffer against dietary acidity, improve hydration at a cellular level, and provide essential dietary minerals.[1]

The challenge, however, lies in the precision of its production. The two most common methods each have significant drawbacks:

1. **Electrolysis (Water Ionizers):** These devices use electricity to split water into acidic and alkaline streams. While they can raise pH, they are expensive, require regular maintenance and descaling, and the resultant water's mineral content is dependent on the minerals *already present* in the source water. If the source water is soft (low in minerals), the ionizer may produce high-pH water that is essentially caustic and unhealthy.[10]
2. **Static Mineral Cartridges/Filters:** These are passive systems where water flows through a cartridge containing minerals that slowly dissolve. They are simpler and cheaper but offer no control or adaptability. As the cartridge depletes, the output pH and mineral concentration gradually decrease. They cannot adjust to changes in the incoming water's chemistry.[11]

The core problem is that both methods are **reactive and non-adaptive**. They lack the intelligence to understand what is in the water and the control to precisely adjust it to a desired, safe, and beneficial state. This can lead to inconsistent quality, potential health risks (if pH is too high or mineral balance is wrong), and wasted minerals.

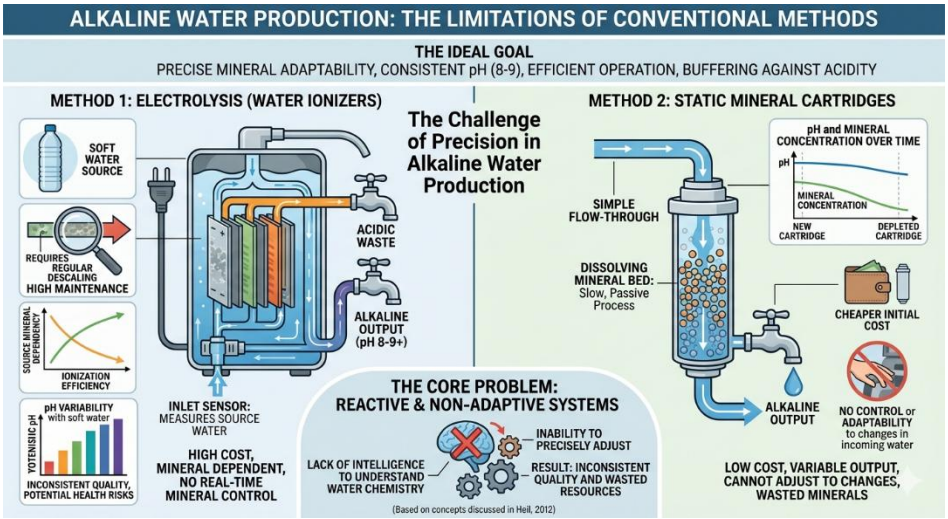



*Figure 1. An overview of the alkaline water production*

## 1.2 Literature Review

- **Health and Properties of Alkaline Water:** Foundational research by Heil [1] (2012) explored the physicochemical properties of alkaline water and its potential physiological effects. This and subsequent studies [8] provide the rationale for why controlled alkalinity and mineralization are desirable, establishing the target parameters (pH, specific minerals) that a smart system must aim for.

*Table 1. Comparison of characteristics between alkaline water drinkers and non-drinkers [17]*

| Variables | Alkaline water drinkers (n = 148) | Non-alkaline water drinkers (n = 156) | Overall (n = 304) | p value* |
|---|---|---|---|---|
| Age (years) | 67.8 ± 6.4 | 68.4 ± 5.5 | 68.1 ± 5.9 | 0.074 |
| Years of education | 10.8 ± 3.6 | 10.6 ± 4.1 | 10.7 ± 3.9 | 0.085 |
| Employed | 23 (15.5) | 18 (11.5) | 41 (13.5) | 0.055 |
| Monthly family income (MYR) | 6542±2525 | 5938±3206 | 6232±2488 | 0.069 |
| Body weight (kg) | 62.9 ± 7.2 | 64.6± 7.0 | 63.8 ± 7.1 | 0.046 |
| BMI (kg/m$^2$) | 25.8 ± 3.5 | 26.1 ± 3.3 | 26.0 ± 3.4 | 0.060 |
| % Body fat | 40.6±5.8 | 44.2 ±4.8 | 42.4 ± 5.6 | 0.045 |
| Physical activity (MET-min/week) | 823±426 | 934±501 | 880 ±469 | 0.058 |
| Insufficient active (<600 MET-min/week) | 66 (44.5) | 77 (49.4) | 143 (47.0) | 0.066 |
| Low active (600–3999 MET-min/week)** | 82 (55.5) | 79 (51.6) | 161 (53.0) | |
| Hand grip strength (kg) | 20.8±2.8 | 15.5±4.1 | 18.1 ± 4.4 | 0.004 |
| Good hand grip | 83 (56.1) | 40 (25.6) | 123 (40.5) | 0.002 |
| Poor hand grip* | 65 (43.9) | 116 (74.4) | 181 (59.5) | |
| Duration of sleep (minutes) | 334±62 | 275± 59 | 304± 67 | 0.008 |
| < 5 hour per day | 48 (32.4) | 89 (57.1) | 137 (45.1) | 0.012 |
| 5–6 hour per day | 74 (50.0) | 52 (33.3) | 126 (41.4) | |
| > 6 hour per day | 26 (17.6) | 15 (9.6) | 41 (13.5) | |
| Sleep Quality Score | 3.7 ± 2.6 | 6.5 ± 3.1 | 5.14 ± 2.4 | 0.009 |
| Poor Sleeper | 81 (54.7) | 114 (73.1) | 195 (64.1) | < .008 |
| Good Sleeper | 67 (45.3) | 42 (26.9) | 109 (35.6) | |
| Dietary Quality Score | 60.9 ± 8.23 | 59.6 ± 6.58 | 60.2 ± 4.9 | 0.120 |
| Dietary Acid Load | 20.4 ± 7.5 | 24.5 ± 3.3 | 22.5 ± 6.08 | 0.072 |
| Presence of Metabolic Syndrome | 61 (41.2) | 84 (53.8) | 145 (47.7) | 0.041 |
| SBP (mmHg) | 136 ±24 | 138±21 | 137±23 | 0.061 |
| DBP (mmHg) | 92 ±11 | 97± 7 | 95 ± 10 | 0.075 |
| Elevated blood pressure | 76 (51.3) | 85 (54.5) | 161(53.0) | 0.058 |
| Waist Circumference (cm) | 77.3 ±11.3 | 85.2±10.1 | 81.3±11.4 | 0.034 |
| Elevated Waist Circumference | 61 (41.2) | 92 (59.0) | 153 (50.3) | 0.038 |
| LDL (mmol/L) | 3.68 ±0.48 | 3.82 ±0.43 | 3.75±0.46 | 0.074 |
| Elevated LDL | 92 (62.1) | 111 (71.2) | 203 (66.8) | 0.157 |
| HDL (mmol/L) | 0.83 ± 0.18 | 0.82 ±0.21 | 0.82 ±0.20 | 0.062 |
| Low HDL | 110 (74.3) | 63 (40.4) | 173 (56.9) | 0.068 |
| TG (mmol/L) | 1.60 ±0.39 | 1.89 ±0.34 | 1.75±0.40 | 0.029 |
| Elevated serum TG | 51 (34.9) | 64 (41.0) | 115 (37.8) | 0.065 |
| FBG (mmol/L) | 4.69 ±1.03 | 5.11 ±1.02 | 4.90 ±1.04 | 0.038 |
| Elevated FBG | 38 (25.7) | 50 (32.1) | 88 (28.9) | 0.056 |
| TG/HDL | 2.03 ± 0.65 | 2.51 ± 0.14 | 2.27 ± 0.2f5 | 0.035 |

- **Existing Alkaline Water Generation Technologies:** The literature describes two main technological approaches. Electrolysis-based ionizers are well-documented in patents and product literature [10]. Their limitations in terms of dependency on source water minerals are a known engineering challenge. Passive mineral cartridge systems are also widely described [11], with their primary drawback being the lack of feedback and control (Kumar & Sharma [3], 2019, in the context of general water quality monitoring, highlights the need for real-time data).

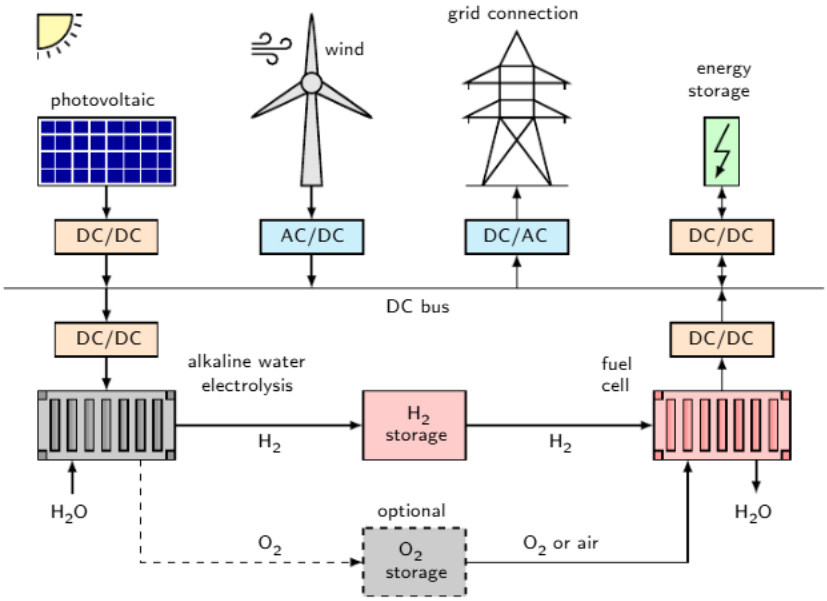



*Figure 2. Schematic process scheme of a hydrogen energy system*

- **IoT and AI in Water Treatment:** The application of IoT for real-time water quality monitoring is a mature field [3] (Kumar & Sharma, 2019). Researchers have successfully deployed sensor networks to track parameters like pH, TDS, and temperature. More recently, the focus has shifted to using this data for intelligent control. Chen et al. [2] (2021) explored AI-enabled mineral balancing in industrial water treatment contexts, using machine learning to optimize the addition of chemicals for processes like scale inhibition. This demonstrates the feasibility of using ML for precise dosing control, but their work is focused on industrial scale and applications, not consumer-level alkaline water generation. Singh & Patel [4] (2020) further discuss the potential of smart water purification using machine learning, but their review is general and does not delve into the specific control problem of automated mineral dosing for alkalinity.

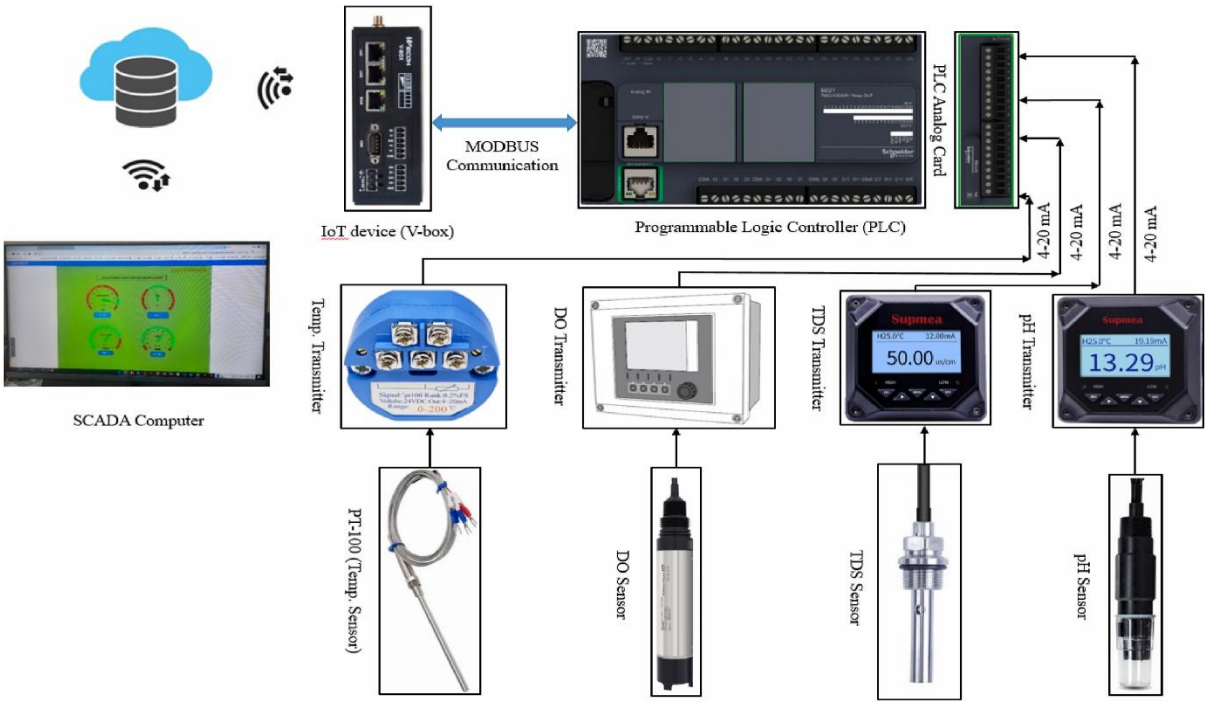



*Figure 3. Block diagram of the system*

- Real-time water quality monitoring approach

In IoT based configuration part, first the IoT (V-box) device has bound through the V-net software.

V-net is an internet-based configuration platform for the IoT device (V-box). To access the V-net first registration has done by setting a unique ID and password for login to the V-net platform. Then configuration has done for real-time data, alarm, and historical data. After completing the configuration for real-time data, alarm and historical data web cloud configuration has done for monitoring and control from the cloud server. Web cloud configuration is a graphical representation for monitoring real time data, historical data acquisition and trend display systems.

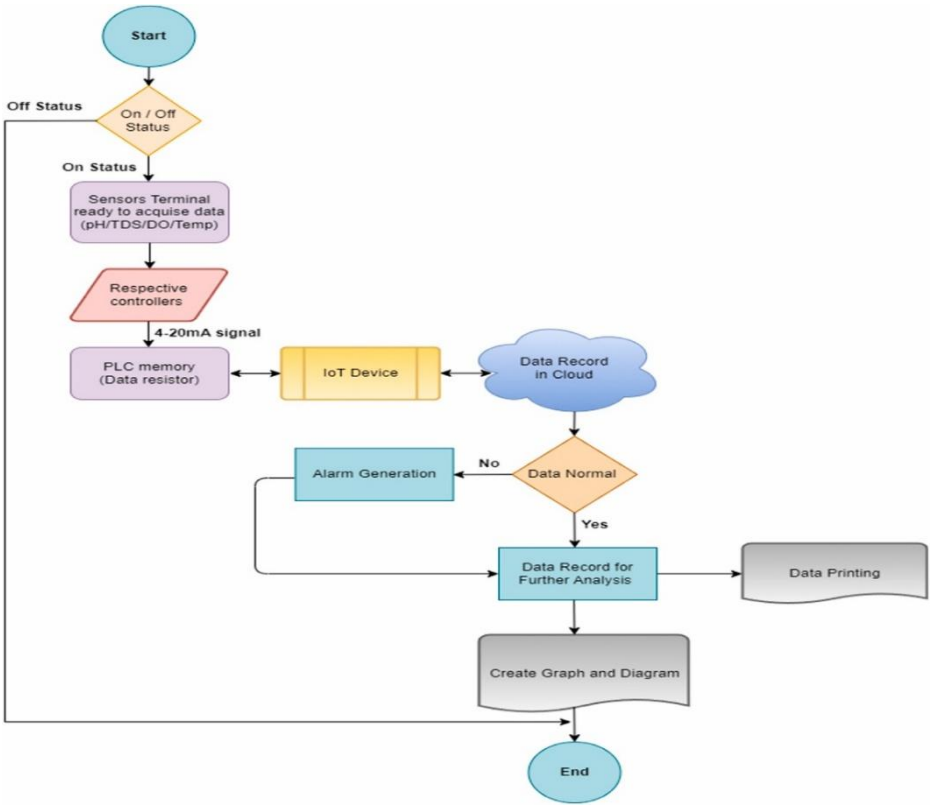



*Figure 4. Process flowchart*

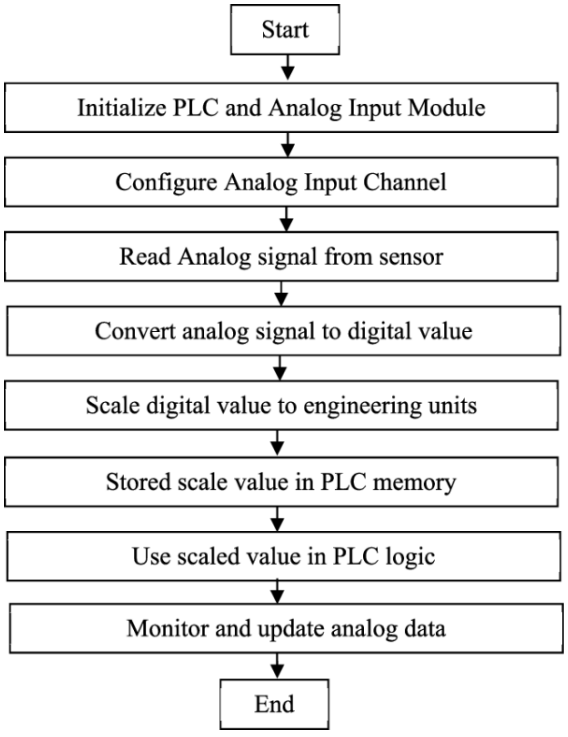



*Figure 5. PLC flowchart*

*Table 2. Illustrates the previous IoT based WTPs systems hardware and software function*

| Description | Used controller and IoT device | Software program | Limitations | Ref. |
|---|---|---|---|---|
| A hybrid machine learning and embedded IoT-based water quality monitoring system. | Arduino microcontroller, ESP8266 Wi-Fi module | Message Queuing Telemetry Transport (MQTT), Proteus 8 professional | Microcontroller based system is not suitable for industrial applications, limited industrial reliability, not designed for harsh environments, limited I/O and expandability | [28] |
| Development of LoRaWAN-based IoT system for water quality monitoring in rural areas. | Arduino microcontroller, LoRaWAN gateway | Fritzing software, Arduino IDE software | | [29] |
| Real-time IoT-powered AI system for monitoring and forecasting of air pollution in industrial environment. | Microcontroller (Arduino MEGA), Narrowband IoT GSM | Machine learning | | [30] |
| Advances in real time smart monitoring of environmental parameters using IoT and sensors. | Aruduino Uno-ATMega 1280, ESP8266 Wi-Fi module, GSM module | C | | [31, 32] |

*Table 3. Configuration for real time data, alarm and accuracy*

| Parameter | Range | Accuracy | Working Temp | Alarm |
|---|---|---|---|---|
| pH (SUP-PH6.0) | 0–14 | ±0.2 | 0–80 °C | Less than 5 and greater than 9 |
| TDS (SUP-TDS210-B) | 0–1000 ppm | ±0.1 | 0–100 °C | Less than 200 ppm and greater than 900 ppm |
| DO (COS61D-AAA1A3) | 0–20 mg/L | ±0.1 | 0–80 °C | Less than 5 mg/L |
| Temperature (PT-100) | 0-200 °C) | ±0.1 | - 50–120 °C | Less than $15^0$ C and greater than $45^0$ C |

## 1.3 Research Gap

The consumer market for alkaline water has experienced explosive growth, driven by purported health benefits and increasing public awareness of water quality. This demand has been met by a proliferation of devices, from basic pitcher filters to sophisticated electronic water ionizers. A critical examination of these products, however, reveals a fundamental and pervasive technological limitation: they are all, without exception, **"open-loop" systems**. They operate on the assumption of consistent input and perfect mechanical function, possessing no intrinsic ability to verify their own performance, adapt to changing conditions, or confirm that they are delivering the promised—and safe—output to the consumer. This represents not merely an incremental design flaw, but a profound gap between consumer expectation and technological reality.

This research addresses this critical void by proposing the first **closed-loop, intelligent control system for consumer alkaline water generation**. The gap is multifaceted and can be deconstructed into three primary dimensions:

**1. The Foundational Gap: The "Open-Loop" Problem and the Illusion of Control.**

The entire consumer alkaline water industry is built upon an "open-loop" control architecture. In terms of engineering, an open-loop system executes a command without using feedback to verify the result. This is precisely how current devices operate:

- **Ionizers:** A user selects a pH setting (e.g., 9.5). The device applies a preset voltage to its electrolysis plates for a predetermined duration, *assuming* this will produce water at the target pH. It has no sensor to measure the actual pH of the output water. If the input water chemistry changes seasonally, if the plates have mineral buildup, or if the flow rate fluctuates, the output pH drifts silently. The user is given an illusion of control, but the device is flying blind.
- **Cartridge Filters (e.g., magnesium-based):** These devices rely on contact time between water and a mineral medium. As the cartridge nears depletion, the mineral release diminishes. The user experiences a gradual, often imperceptible, decline in water alkalinity and has no reliable indicator of when the cartridge is exhausted, other than a vague taste difference or a simple timer.

This "open-loop" paradigm creates two critical failures: **inconsistency** (the device fails to reliably deliver its core function) and **unsafety** (if a unit malfunctions or a cartridge is depleted, it cannot alert the user, potentially delivering water that is not only non-alkaline but also chemically imbalanced). The foundational gap is the absence of a feedback mechanism, the missing sensor that closes the loop and transforms a guessing appliance into a precision instrument.

**2. The Control Gap: From Static Assumptions to Dynamic, Predictive Actuation.**

The logical progression from an open-loop system is a closed-loop one, where a sensor measures the output and a controller adjusts the actuator to maintain the setpoint. However, a simple feedback loop (e.g., "if pH is low, turn on the pump more") is reactive and can be unstable. The true frontier, and the location of the second gap, lies in **predictive, feed-forward control using machine learning**.

There is currently no evidence of a low-cost, integrated system that applies ML to the specific challenge of real-time mineral dosing for consumer water. The complexity lies in the non-linear relationship between multiple variables:

- **Inputs:** Source water pH, Total Dissolved Solids (TDS), temperature, flow rate.
- **Actuators:** Pump speed and duration for a mineral slurry, or variable voltage for electrolysis.
- **Output:** Target pH and mineral concentration.

An ML model presents the ideal solution to this multivariate control problem. It can be trained to learn the complex transfer function of the system, enabling it to predict, in real-time, the precise actuator commands required to achieve a target water profile, *before* the output deviates. This shifts the control paradigm from reactive correction to **predictive precision**. The gap is not just the absence of a closed loop, but the absence of an *intelligent* loop that anticipates and adapts.

**3. The Human-Centric Gap: The Absence of Personalization and Adaptation.**

The final dimension of the gap is the complete lack of user-centric intelligence in existing products. Current devices are static, one-size-fits-all appliances. They cannot account for individual preferences, specific health goals, or the long-term evolution of the system itself. This creates a third gap: the absence of a **personalized and adaptive water experience**.

A truly intelligent system would transcend simple setpoint maintenance and offer:

- **User-Defined Targeting:** A system that allows a user to specify a desired water profile (e.g., "pH 9.0 with high magnesium") and autonomously works to achieve that precise output, regardless of input variations.
- **Adaptive Learning:** A system that employs a machine learning model which continuously improves over time. It could learn from the long-term effects of component wear (e.g., pump degradation, sensor drift) and adjust its control algorithm to compensate, maintaining consistent performance without recalibration.

- **Context-Aware Intelligence:** A system that could, for example, detect an anomaly in the input water (e.g., a spike in source water TDS) and automatically adjust its dosing strategy, or alert the user that the incoming water quality is outside the unit's optimal operating range.

This research directly confronts this tripartite gap. It proposes the development of an **intelligent, closed-loop control module** that integrates real-time multi-parameter sensing with an on-device machine learning model. This model will bridge the gap between measured input/output and precise actuation, creating a system that is not only predictive and adaptive but also capable of learning and personalizing the water generation process for the end-user. This moves the consumer device from a passive, open-loop appliance to an active, intelligent, and reliable "water sommelier" that can guarantee the quality and consistency of every drop it produces.

# 2. OBJECTIVE

## 2.1 Main Objective

To design, develop, and validate an **intelligent, closed-loop alkaline water generation module** that integrates IoT-based real-time sensing with a machine learning-driven control algorithm to automatically and precisely maintain a user-defined, safe, and mineral-balanced alkaline water profile.

## 2.2 Specific Objectives

1. **Multi-Parameter Sensing and Data Acquisition:** To design and implement a robust data acquisition system using an ESP32 microcontroller and a suite of sensors (pH, TDS, temperature, and potentially specific ion-selective electrodes for calcium/magnesium) to continuously monitor the water quality before and after mineralization.
2. **Precision Automated Dosing Mechanism:** To design and fabricate a precise and reliable automated dosing system, likely using multiple peristaltic pumps, capable of injecting micro-liter quantities of concentrated mineral solutions (e.g., calcium carbonate, magnesium sulfate, potassium bicarbonate) into the water stream based on digital control signals.
3. **Machine Learning Model for Dosage Prediction:** To develop and train a supervised machine learning regression model (e.g., ANN, SVR, or Gradient Boosting) using a curated dataset. The model will learn the complex, non-linear relationship between input water parameters (source pH, TDS), desired output parameters (target pH, mineral levels), and the required dosage of each mineral solution.

4. **Closed-Loop Control System Integration:** To integrate the sensors, ML model, and dosing mechanism into a single, cohesive closed-loop system. The ESP32 will read sensor data, pass it to the ML model (which may run on the ESP32 itself or a connected Raspberry Pi/cloud server), receive the dosage command, and actuate the pumps, accordingly, creating a continuous feedback loop.
5. **User Interface and System Validation:** To develop a simple user interface (e.g., a mobile app or local web dashboard) that allows users to set their desired alkaline water profile (target pH range) and monitor system performance. The complete system will be rigorously validated to assess its accuracy in maintaining the target pH, its response time to disturbances, and its overall reliability.

# 3. METHODOLOGY

The development will follow an iterative, systems engineering approach.

- **Phase 1: Hardware Design and Component Selection**
    - **Sensor Array:** An ESP32 will interface with an analog pH sensor (e.g., Atlas Scientific or Gravity), a TDS sensor, and a DS18B20 temperature sensor (for compensation) [15]. For more advanced prototyping, ion-selective electrodes for calcium and magnesium may be considered.
    - **Dosing Mechanism:** A set of miniaturized peristaltic pumps (3-5 units) will be selected. Peristaltic pumps are ideal as they are precise, self-priming, and keep the mineral solutions contained, preventing contamination [6]. Each pump will be dedicated to a specific mineral solution (Calcium, Magnesium, Potassium).
    - **Mixing and Flow Cell:** A small, continuous-flow mixing chamber will be designed to ensure the injected minerals are thoroughly mixed with the water before reaching the output sensor. A custom 3D-printed flow cell will house the sensors to ensure consistent readings.
- **Phase 2: Firmware and Data Acquisition**
    - **ESP32 Firmware (Arduino IDE/PlatformIO):** Code will be written to:
        - Read and calibrate sensor data at a high frequency (e.g., every second).
        - Implement temperature compensation for pH and TDS readings.
        - Control the peristaltic pumps via digital outputs and motor drivers (or simple transistors/relays), using PWM for finer flow control if needed.

        - Establish a Wi-Fi connection and implement a communication protocol (e.g., MQTT or HTTP) to send sensor data to and receive dosage commands from a central processor (which could be the cloud or a local Raspberry Pi).

- **Phase 3: Machine Learning Model Development**
    - **Dataset Creation:** An automated data collection rig will be built. The system will run through thousands of cycles, varying the "source" water (by mixing tap and distilled water) and the mineral pump activation times. For each combination, the input parameters (source pH, source TDS) and the resulting output parameters (output pH, output TDS) will be logged. This creates a labeled dataset where the features are [source_pH, source_TDS, pump1_time, pump2_time, ...] and the target is [output_pH, output_TDS].
    - **Model Training (Python - scikit-learn/TensorFlow):** The dataset will be used to train a regression model. The goal is to create a model that can predict the required pump activation times (the "dosage") given the current source water parameters and the desired target parameters. Models like Multi-Output Regression with Random Forests, or a simple Neural Network, will be explored and evaluated based on prediction error (MAE) for the final pH [2],[9].
- **Phase 4: Closed-Loop Integration and Control Logic**
    - The final control logic on the ESP32 will be:
        1. Read source water sensor data (pre-dosing).
        2. Send this data, along with the user's target pH (e.g., 8.5), to the ML model (via a local API if on a Raspberry Pi, or to a cloud endpoint).
        3. Receive the predicted pump activation times back from the model.
        4. Activate the peristaltic pumps for the calculated durations.
        5. Read the output sensor data (post-dosing) to verify the result. This verification data can be used for logging and for future model retraining/improvement.
- **Phase 5: User Interface and Validation**
    - **Mobile/Web Dashboard:** A simple dashboard (e.g., using Flutter for mobile or react for web) will allow the user to set the target pH and view real-time sensor data and system status.
    - **Validation:** The complete system will be tested by attempting to maintain a constant target pH while the source water is deliberately varied (e.g., by switching between tap water and a more mineral-rich source). The system's settling time, overshoot, and steady-state error will be measured and documented.

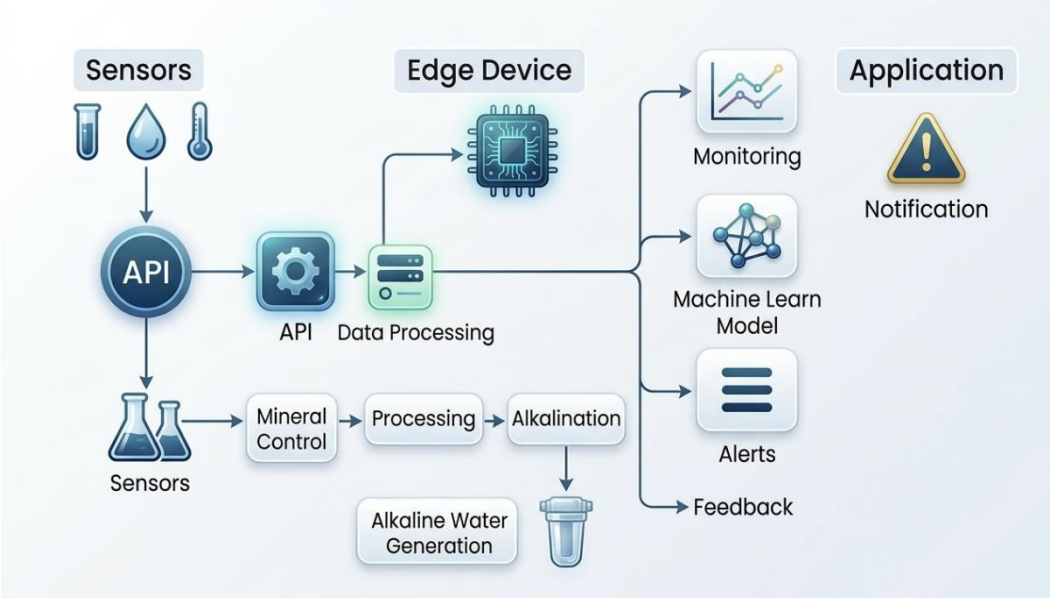



*Figure 6. Overall system architecture for smart water generation using IoT and AI*

# 4. HIGH-LEVEL SYSTEM ARCHITECTURE

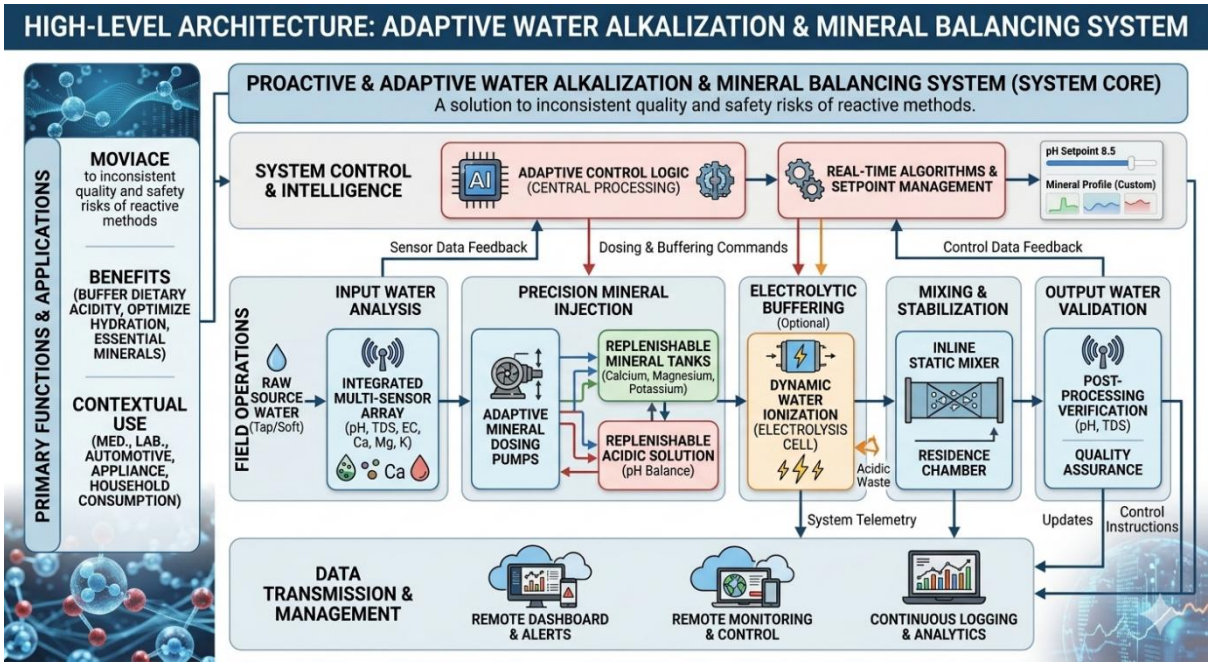



*Figure 7. High-level system architecture*

# 5. USER REQUIREMENTS

The system will be designed with the following requirements in mind:

**Functional Requirements:**

- **Monitoring -** The system shall continuously monitor and display the pH and TDS of both the incoming source water and the outgoing alkaline water.
- **Dosing Control -** The system shall automatically inject precise amounts of at least three mineral solutions (Calcium, Magnesium, Potassium) to achieve the target water profile.
- **User Target Setting -** The user shall be able to set a desired target pH for the output alkaline water within a safe range (e.g., 8.0 to 9.5) via the user interface [7][14].
- **Alerts -** The system shall generate alerts for abnormal conditions, such as when a mineral cartridge is empty, when sensors require calibration, or if the output pH falls outside a safe tolerance band.
- **Data Logging -** The system shall log historical water quality data and mineral usage for user review and system analysis.

**Non-Functional Requirements:**

- **Accuracy -** The system must maintain the output water pH within **±0.15** of the user's target value under steady-state conditions [14].
- **Response Time -** The system should achieve the target pH value within a reasonable time frame (e.g., within 2-3 minutes of a change in source water) without significant overshoot.
- **Reliability -** The dosing mechanism must be reliable and precise, with minimal variation in the amount of mineral dispensed per activation [6].
- **Safety -** The system must include a failsafe mechanism to immediately stop mineral dosing and alert the user if any sensor reading indicates an unsafe condition (e.g., post-dosing pH > 10) [7].
- **Maintainability -** Mineral cartridges should be easily replaceable by the user, and the system should provide clear notifications when replacement is due.
- **Usability -** The user interface must be intuitive, presenting complex data in a simple, understandable format (e.g., gauges, color-coded indicators)

# 6. COMMERCIALIZATION PLAN

This intelligent alkaline water module has significant market potential as a standalone product or as an integrated component.

- **Target Market Segments:**
    1. **Health and Wellness Consumers:** The primary market is individuals and families actively seeking products that promise health benefits [1],[8]. This includes users of existing alkaline water products who are dissatisfied with their inconsistency.
    2. **High-End Residential Market:** The module can be positioned as a premium under-sink or countertop appliance for smart homes, appealing to tech-savvy homeowners who value both health and automation.
    3. **Commercial and Hospitality:** Wellness centers, gyms, yoga studios, upscale offices, and hotels could offer "intelligent alkaline water" as a premium amenity to their clients and employees.
    4. **OEM Integration:** The module can be designed as a "drop-in" component for manufacturers of existing water purifiers and dispensers, allowing them to instantly add a high-value "smart alkaline" feature to their product lines.
- **Market Strategy:**
    - **Direct-to-Consumer (DTC):** Launch via a dedicated website and e-commerce platforms, using content marketing (blog posts, videos explaining the science and technology) and social media advertising targeting health and wellness communities.
    - **Partnerships:** Form partnerships with high-end kitchen appliance retailers, interior designers, and wellness product distributors.
    - **B2B Licensing:** Actively pitch the module as a licensable technology to major water purification companies in Sri Lanka and the region.
- **Revenue Model:**
    - **Hardware Sales:** Primary revenue from the sale of the complete alkaline water generation unit.
    - **Recurring Revenue (Consumables):** A key, sustainable revenue stream from the sale of proprietary, refillable mineral cartridges. This creates a long-term customer relationship.
    - **Premium Software Features:** Offer a subscription tier for the mobile app that provides advanced analytics, historical trend reports, and personalized health insights based on consumption data.

- **Competitive Advantage:** The key differentiator is **precision and intelligence**. Unlike "dumb" ionizers or passive filters, this system guarantees consistent results, adapts to your water, and provides complete transparency and control through its smart features. It transforms a simple appliance into a trusted health device.

# 7. BUDGET AND JUSTIFICATION

*Table 4. Budget and justification*

| Item | Component/Description | Estimated Cost (LKR) | Justification |
|---|---|---|---|
| **1. Dosing Mechanism** | 3x Mini Peristaltic Pumps, Silicon Tubing, Check Valves | 3,000 | Precision pumps are essential for accurate mineral dosing. Using three pumps allows for multi-mineral control. |
| **2. Fluidics & Housing** | 3D Printed Flow Cell/Mixing Chamber, Mineral Solution Reservoirs (3x), Tubing, Fittings | 4,500 | For containing mineral solutions and ensuring proper mixing and sensor placement. |
| **3. Mineral Solutions** | Food-grade Calcium, Magnesium, Potassium compounds for creating concentrated solutions | 3,000 | The "consumables" for creating the alkaline water, needed for prototyping and testing. |
| **4. Misc. Hardware** | Breadboard, Jumper Wires, Motor Drivers/Relays, PCB (optional) | 2,000 | For assembling and connecting all electronic components. |
| **TOTAL ESTIMATED BUDGET** | | **12,500** | |

# 8. WORK BREAKDOWN STRUCTURE (WBS)




*Figure 8. Work breakdown structure*

# 9. GANTT CHART

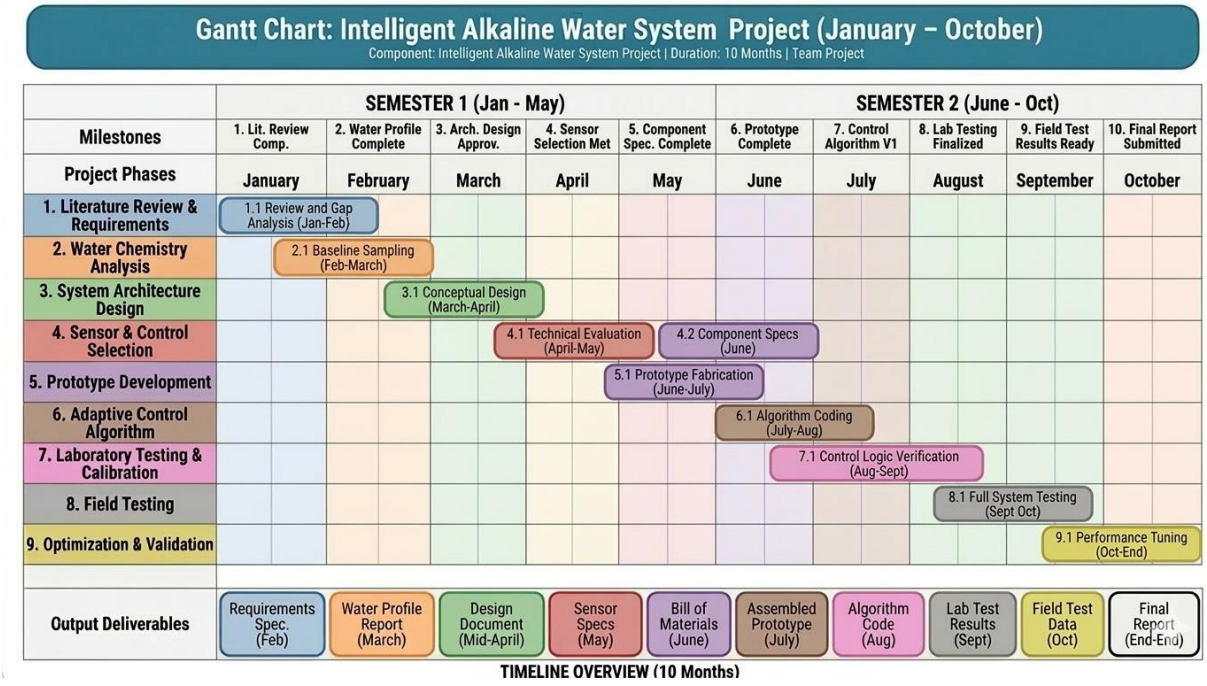



*Figure 9. Gantt chart*

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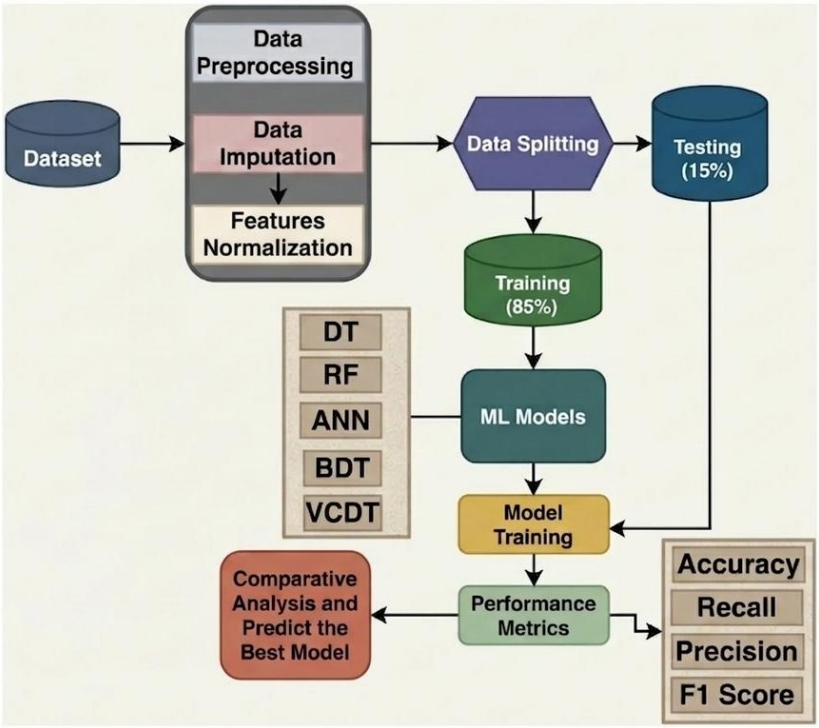
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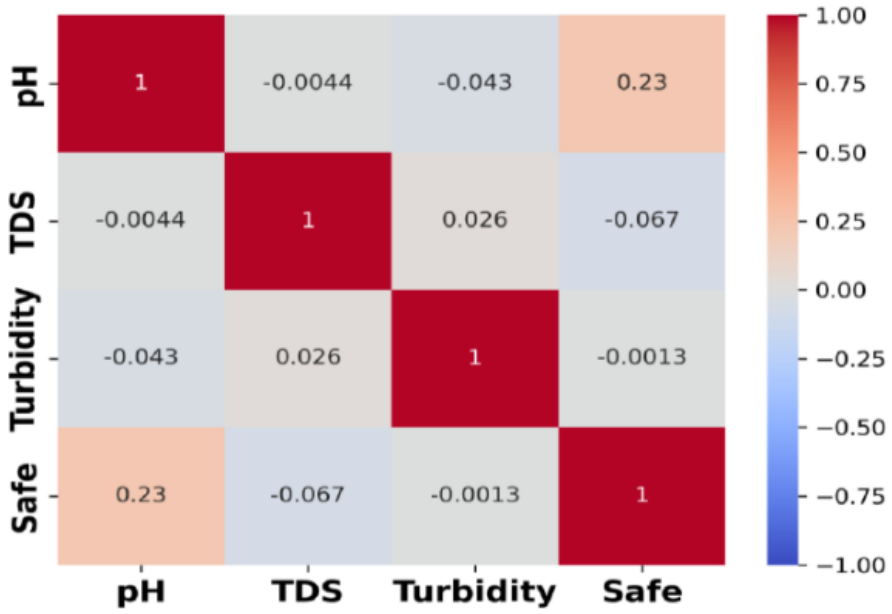
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# 11.APPENDICES




*Appendix A: Machine learning working procedure*




*Appendix B: Pearson correlation matrix analysis*

| Model | Accuracy (%) | Precision (%) | Recall (%) | F1 Score (%) |
|---|---|---|---|---|
| ANN | 92.66 | 93.33 | 92.87 | 92.65 |
| Random Forest | 84.28 | 85.19 | 84.53 | 84.23 |
| Bagging Decision Tree | 83.02 | 84.01 | 83.29 | 82.96 |
| Voting Classifier | 83.23 | 84.49 | 83.53 | 83.15 |
| Decision Tree | 75.26 | 75.41 | 75.37 | 75.26 |

*Appendix C: Performance metrics for the machine learning algorithms utilized in this study's dataset*

### A-Organic Beverages

| Content/Serving | Alkali (mEq) | Na (mg) | Calories | Cost per month to achieve 30 mEq/day |
|---|---|---|---|---|
| 1. Simply® Orange, Pulp[1] | 15 | 0 | 110 | $37 |
| 2. Tropicana® 50 OJ, Some Pulp [1] | 14 | 10 | 50 | $34 |
| 3. Tropicana® OJ, No Pulp[1] | 13 | 0 | 110 | $44 |
| 4. Trop50® OJ, No Pulp[1] | 13 | 10 | 50 | $44 |
| 6. Kroger® Low Cal OJ[1] | 11 | 5 | 60 | $43 |
| 7. Simply® Grapefruit Juice[2] | 9 | 10 | 100 | $59 |
| 8. Mott's® Apple Juice* | 5 | 30 | 120 | $78 |
| 9. Lakewood® Pure Cranberry Juice[2] | 4 | 15 | 120 | $720 |
| 10. Mott's® Light Apple Juice* | 4 | 30 | 50 | $493 |
| 11. Taste of Nirvana Coconut Water[3] | 3 | 48 | 36 | $1295 |
| 12. Welch's® White Grape Juice* | 2 | 15 | 140 | $207 |
| 13. MinMaid® Lite Lemonade[4] | 2 | 15 | 0 | $191 |
| 14. MinMaid® Lemonade[4] | 1 | 35 | 140 | $135 |
| 15.Ocean Spray® Cranberry Juice Cocktail[2] | 1 | 5 | 110 | $260 |
| 16. GTS Gingerade® Kombucha* | <1 | 10 | 50 | $2532 |
| 17.Braggs® Apple Cider Vinegar[2] | <1 ◦ | 0 ◦ | 0 ◦ | $4161◦ |
| 17. Alkaline Waters (high pH, low mineral)* | <1 | 0 | 0 | N/A |

### B-Sodas

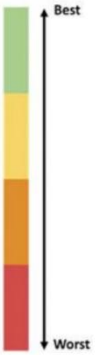
| Content/Serving | Alkali (mEq) | Na (mg) | Calories | Cost per month to achieve 30 mEq/day |
|---|---|---|---|---|
| 1.Diet Sunkist® Orange[6] | 2 | 110 | 0 | $150 |
| 2. Diet 7 Up®[6] | 2 | 45 | 0 | $148 |
| 3. Diet Canada Dry®[6] | 2 | 120 | 0 | $158 |
| 4. Sierra Mist® Free[6] | <2 | 35 | 0 | $659 |
| 5. Sprite® Zero[6] | <2 | 35 | 0 | $194 |
| 6. Diet Orange Crush®[6] | <2 | 75 | 0 | $230 |
| 7. Fresca®[6] | <2 | 50 | 0 | $227 |
| 8. Diet Mt Dew®[6] | <2 | 70 | 0 | $270 |
| 9. Fanta® Orange Zero[6] | 1 | 35 | 0 | $416 |
| 10. Diet Squirt®[6] | <1 | 55 | 0 | $416 |
| 11. Diet Mug® Root Beer[6] | <1 | 65 | 0 | $716 |
| 13. Diet Pepsi® No Caffeine[6] | <1 | 35 | 0 | $15,406 |
| 14. Coke® Zero[6] | <1 | 40 | 0 | $15,406 |

### C-Synthetic Beverages

| Content/Serving | Alkali (mEq) | Na (mg) | Calories | Cost per month to achieve 30 mEq/day |
|---|---|---|---|---|
| 1. Crystal Light Lemonade On the Go[1] | 8 ‡ | 75 ‡ | 10 ‡ | $62 ‡ |
| 2. Shaklee Performance® Lime[5] | 5 | 130 | 100 | $212 |
| 3. Crystal Light Orange On the Go[1] | 2 ‡ | 0 ‡ | 5 ‡ | $273 ‡ |
| 4. Gatorade® Lime* | 2 | 107 | 53 | $205 |
| 5. Gatorade® Orange* | 1 | 107 | 53 | $288 |

### D-Over-the-Counter and Medical Food Supplements

| Content/Serving | Alkali (mEq) | Na (mg) | Calories | Cost per month to achieve 30 mEq/day |
|---|---|---|---|---|
| 1. Arm & Hammer® Baking Soda* | 17 † | 315 † | 0 † | < $1 † |
| 2. Moonstone® Powder[7] | 16 ‡ | 109 ‡ | 15 ‡ | $96 ‡ |
| 3. UrocitK* | 15 Δ | 0 Δ | 0 Δ | $239 Δ |
| 4. KSP tabs™[7] | 14 ‡ | 224 ‡ | 10 ‡ | $57 ‡ |
| 5. NOW® Potassium Citrate[7] | 12 | 1.0 | 0 | $5 |
| 6. LithoBalance™ [2,7] | 12 | 15 | 4 | $43 |
| 7. KSP™ Very Berry Tablets[2] | 9 | 219 | 10 | $60 |
| 8. KSP™ Key Lime tablets[2] | 9 | 223 | 10 | $62 |
| 9. TheraLith XR® [7] | 7 Δ | 4 Δ | 0 Δ | $51 Δ |
| 10. Litholyte® Powder[7] | 7 ‡ | 67 ‡ | 0 ‡ | $68 ‡ |
| 11. Litholyte® Coffee[7] | 6 | 28 | 0 | $146 |
| 12. Kidney COP® [7] | 3 Δ | <1 Δ | 0 Δ | $116 Δ |
| 13. Horbäach® Potassium Citrate[7] | 2 Δ | <1 Δ | 0 Δ | $205 Δ |
| 14. Stone Breaker[2] | 0 | 0 | 0 | N/A |




**Standard serving** = 8 oz cup = 240 mL
**† serving** = ¼ Teaspoon
**◦ serving** = 1 Tablespoon
**‡ serving** = 16 oz = 480 mL
**Δ serving** = tablet/capsule
***** = Original Data



*Appendix D: Alkali, Sodium, Calorie, and Cost Comparison of Common Drinks and Supplements*